KW130 SDR Tuner

Evaluation Kit (EVK) v3.0 User Guide

Applicable: KW130 (QFN24 4×4)

Freq: 30 MHz – 2.6 GHz

BW: up to 20 MHz

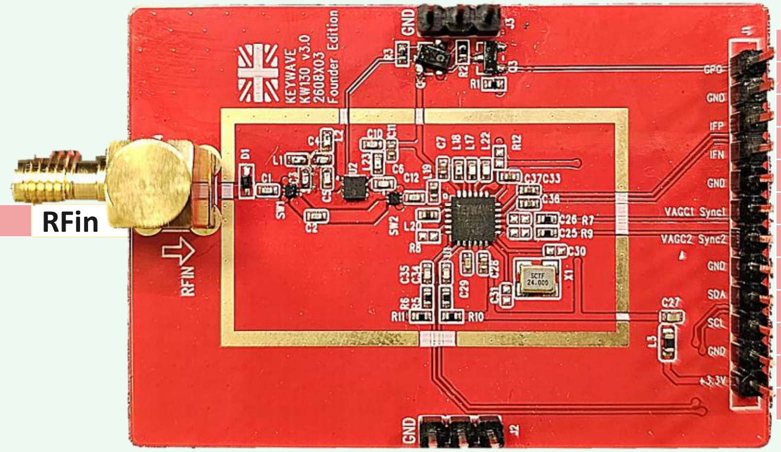
1. Overview

KW130 is a worldwide SDR silicon tuner that eliminates external Baluns and SAW filters. It integrates LNA, smart detector, AGC and selective channel functionality, targeting SDR receivers, DVB, aircraft listening / ADS-B, FM/RDS, and spectrum-analyser front-ends.

The v3.0 EVK include the front LNA with GPIO control with/bypass KW130.

LNA (switch) | AGC integrated | I²C control | 3.3 V single-supply

2. Pinout



0	RFin
1	GPO
2	GND
3	OUTP
4	OUTN
5	GPO
6	SYNC1
7	SYNC2
8	GND
9	SDA
10	SCL
11	GND
12	VDD3V3

● key signal ● standard I/O · NC = not connected

3. EVK Power

Single 3.3V

Recommended operating conditions

VCC = 3.0 V (min) / 3.3 V (typ) / 3.6 V (max)

ICC ≈ 176/225 mA (front LNA off/on) · 4 mA (standby)

TA = −25 °C to +85 °C

Input ripple < 100 mVpp

4. I²C Address Select (R8 on EVK)

R8 value	NC (default)	75 kΩ	22 kΩ	0 Ω
I²C addr	0xF2 ← EVK	0xB2	0x72	0x32

2. Power & Quick Start — FTDI IIC Carrier Board

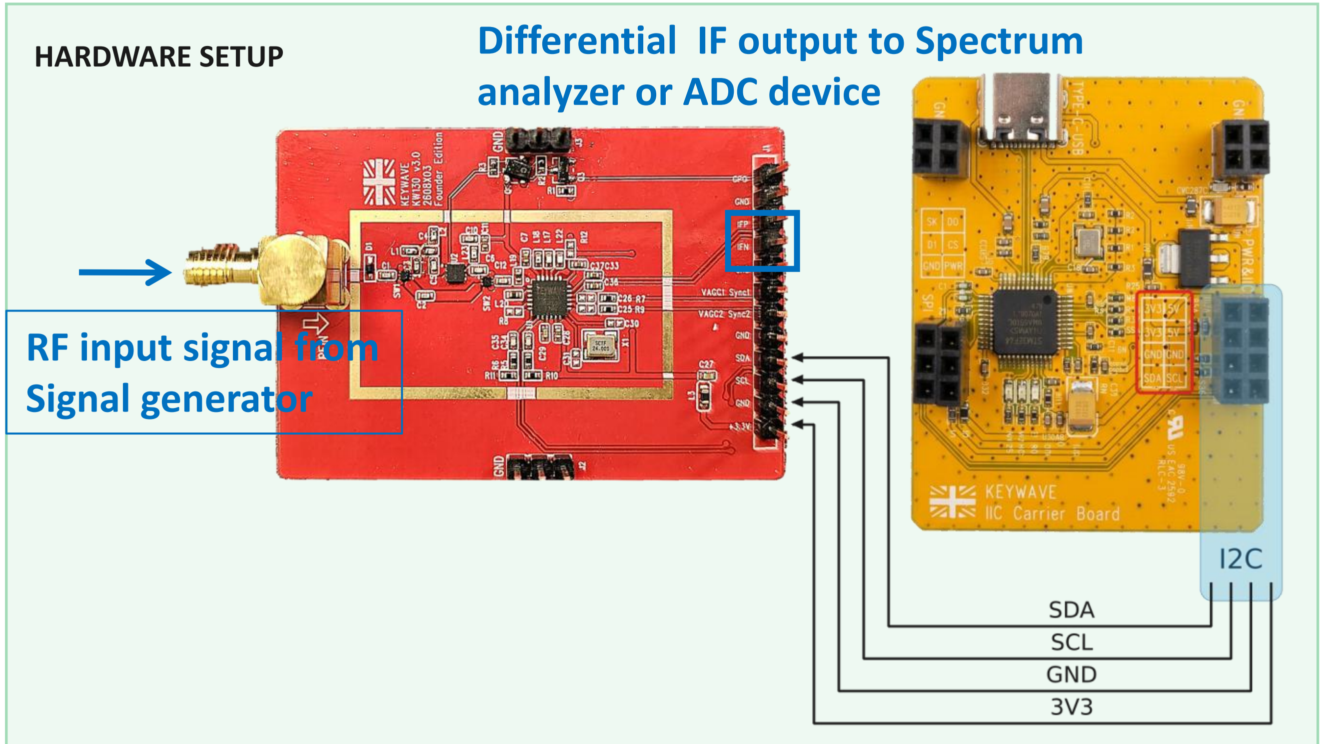
Overview

The KW130 EVK includes a FTDI I2C Carrier Board (yellow PCB) that provides USB-to-I2C bridge functionality, enabling easy connection to a host PC for evaluation and configuration of the KW130 SDR Tuner module.
User can use KW130_SDR_GUI to quick evaluated KW130 Performance through the Carrier Board.

Hardware Setup

- Connect KW130 module to FTDI Carrier Board via the I^{2}C header using the 4-wire cable (VCC, GND, SDA, SCL)
- Connect the FTDI Carrier Board to your PC via the USB port.
- Attach an SMA antenna to the RF connector on the KW130 module
- Measurement the IF output from (IFP/IFN) pin

HARDWARE SETUP



KW130 SDR Tuner + FTDI IIC Carrier Board

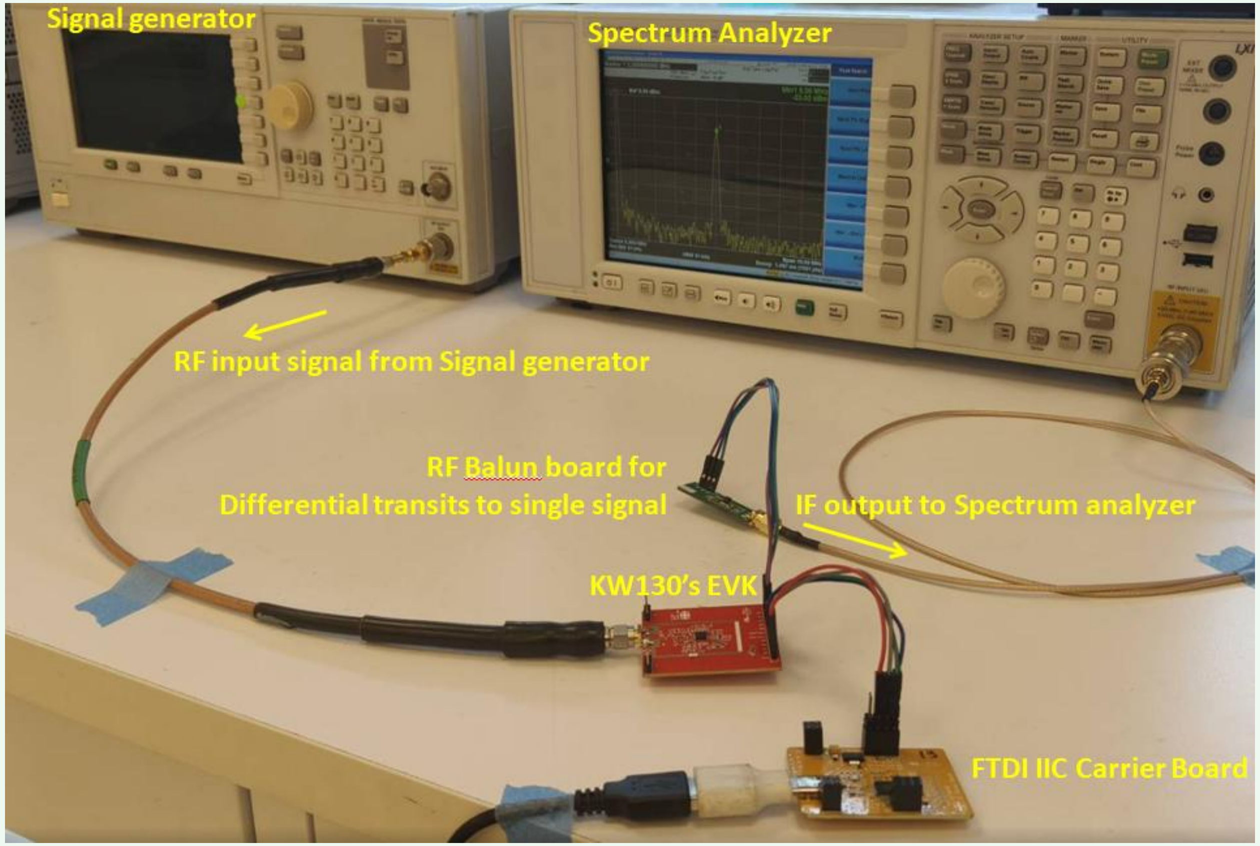
3. Measurement IF output

4.1 KW130's RF input connect to Signal generator with RF cable (Fig4.1)
Set frequency with appropriate power level (ex. 1050M, -60dBm)

4.2 KW130's IF output connect to Spectrum analyzer (Fig4-1)
Observe IF frequency with Spectrum analyzer (ex. "1-10MHz" , RBW=10kHz)

4.3 Observe a CW tone of IF from spectrum analyzer (Fig4-2)

Fig 4.1 KW130's RF input connect to Signal generator & KW130's IF output connect to Spectrum analyzer



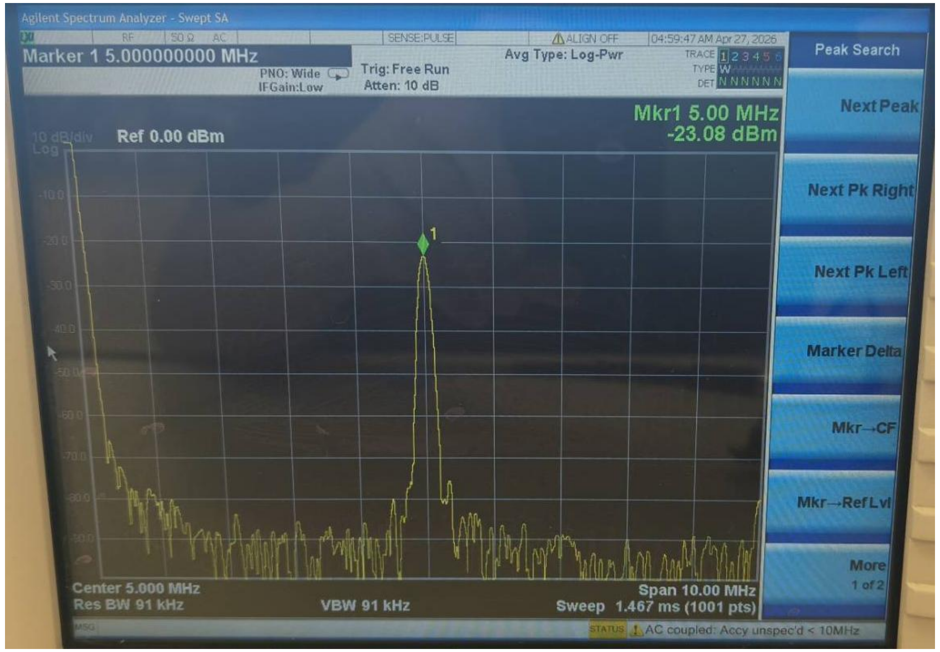
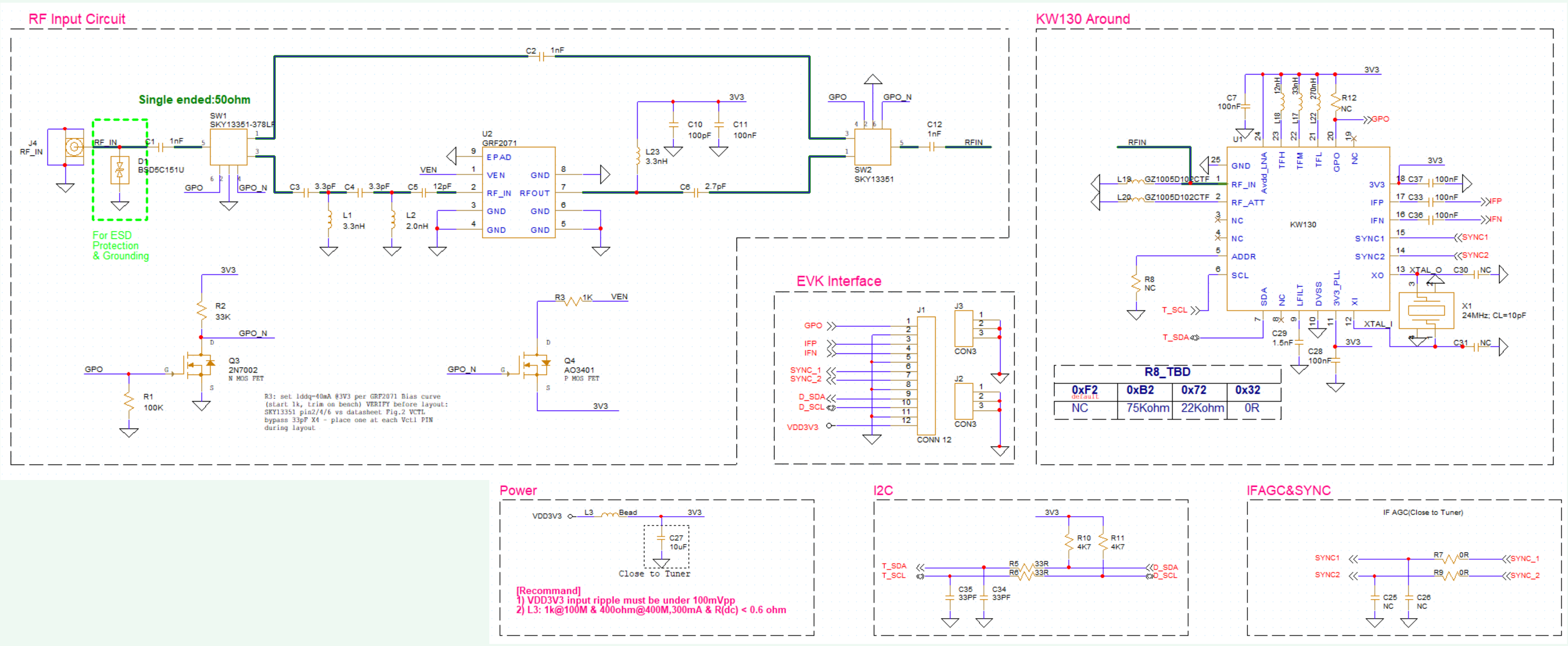


Fig 4.2 Observe a CW tone of IF from spectrum analyzer

EVK control tool operation Guide & Measurement IF output



EVK Design Schematic